2.12 Maclaurin Series - Part 2

Worked Solutions

1 a)

$$\begin{aligned} f(x) &= \sin x, \\ f'(x) &= \cos x, \\ f''(x) &= -\sin x, \\ f^{(3)}(x) &= -\cos x, \\ f^{(4)}(x) &= \sin x. \end{aligned}$$

1 b)

$$\begin{aligned} f(x) &= \cos x, \\ f'(x) &= -\sin x, \\ f''(x) &= -\cos x, \\ f^{(3)}(x) &= \sin x, \\ f^{(4)}(x) &= \cos x. \end{aligned}$$

1 c)

$$f(x) = e^x, \quad f'(x) = e^x, \quad f''(x) = e^x, \quad f^{(3)}(x) = e^x, \quad f^{(4)}(x) = e^x.$$

1 d)

$$\begin{aligned} f(x) &= \ln(1+x), \\ f'(x) &= (1+x)^{-1}, \\ f''(x) &= -(1+x)^{-2}, \\ f^{(3)}(x) &= 2(1+x)^{-3}, \\ f^{(4)}(x) &= -6(1+x)^{-4}. \end{aligned}$$

1 e)

$$f(x) = (1+x)^n, \quad f'(x) = n(1+x)^{n-1}, \quad f''(x) = n(n-1)(1+x)^{n-2},$$
$$f^{(3)}(x) = n(n-1)(n-2)(1+x)^{n-3}, \quad f^{(4)}(x) = n(n-1)(n-2)(n-3)(1+x)^{n-4}.$$

2 a)

$$f(0) = 0, \ f'(0) = 1, \ f''(0) = 0, \ f^{(3)}(0) = -1, \ f^{(4)}(0) = 0.$$
$$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f^{(3)}(0)}{3!}x^3 + \frac{f^{(4)}(0)}{4!}x^4 + \cdots + \frac{f^{(r)}(0)}{r!}x^r.$$

So $\sin x = 0 + x + 0 - \frac{1}{6}x^3 + 0 + \cdots$, hence $\sin x \approx x - \frac{1}{6}x^3$.

2 b)

$$f(0) = 1,\ f'(0) = 0,\ f''(0) = -1,\ f^{(3)}(0) = 0,\ f^{(4)}(0) = 1.$$

$$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f^{(3)}(0)}{3!}x^3 + \frac{f^{(4)}(0)}{4!}x^4 + \cdots + \frac{f^{(r)}(0)}{r!}x^r.$$

So $\cos x = 1 + 0 - \frac{1}{2}x^2 + 0 + \frac{1}{24}x^4 + \cdots$, hence $\cos x \approx 1 - \frac{1}{2}x^2 + \frac{1}{24}x^4$.

2 c)

$$f(0) = 1,\ f'(0) = 1,\ f''(0) = 1,\ f^{(3)}(0) = 1,\ f^{(4)}(0) = 1.$$

$$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f^{(3)}(0)}{3!}x^3 + \frac{f^{(4)}(0)}{4!}x^4 + \cdots + \frac{f^{(r)}(0)}{r!}x^r.$$

So $e^x = 1 + x + \frac{1}{2}x^2 + \frac{1}{6}x^3 + \frac{1}{24}x^4 + \cdots$, hence $e^x \approx 1 + x + \frac{1}{2}x^2 + \frac{1}{6}x^3 + \frac{1}{24}x^4$.

2 d)

$$f(0) = 0,\ f'(0) = 1,\ f''(0) = -1,\ f^{(3)}(0) = 2,\ f^{(4)}(0) = -6.$$

$$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f^{(3)}(0)}{3!}x^3 + \frac{f^{(4)}(0)}{4!}x^4 + \cdots + \frac{f^{(r)}(0)}{r!}x^r.$$

So $\ln(1 + x) = 0 + x - \frac{1}{2}x^2 + \frac{2}{6}x^3 - \frac{6}{24}x^4 + \cdots$, hence $\ln(1 + x) \approx x - \frac{1}{2}x^2 + \frac{1}{3}x^3 - \frac{1}{4}x^4$.

2 e)

$$f(0) = 1,\ f'(0) = n,\ f''(0) = n(n-1),\ f^{(3)}(0) = n(n-1)(n-2),\ f^{(4)}(0) = n(n-1)(n-2)(n-3).$$

$$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f^{(3)}(0)}{3!}x^3 + \frac{f^{(4)}(0)}{4!}x^4 + \cdots + \frac{f^{(r)}(0)}{r!}x^r.$$

So

$$(1 + x)^n = 1 + x + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \frac{n(n-1)(n-2)(n-3)}{4!}x^4 + \cdots.$$

Therefore $(1 + x)^n \approx 1 + x + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \frac{n(n-1)(n-2)(n-3)}{4!}x^4$.

3)

Let $f(x) = \ln x$, then $f(0)$ is undefined. Additionally, $f'(x) = \frac{1}{x}$, and so $f'(0)$ does not exist for $f(x) = \ln x$ (and the higher derivatives also do not exist at $x = 0$). Therefore the Maclaurin series cannot be found.

4 a)

$$f(x) = e^{5x}, \quad f'(x) = 5e^{5x}, \quad f''(x) = 25e^{5x}.$$

So $f(0) = 1$, $f'(0) = 5$, $f''(0) = 25$. Using $f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f^{(3)}(0)}{3!}x^3 + \frac{f^{(4)}(0)}{4!}x^4 + \frac{f^{(5)}(0)}{5!}x^5 + \cdots + \frac{f^{(r)}(0)}{r!}x^r$,

$$e^{5x} = 1 + 5x + \frac{25}{2}x^2 + \cdots, \qquad e^{5x} \approx 1 + 5x + \frac{25}{2}x^2, \quad \text{valid for all } x.$$

4 b)

$$f(x) = \sin 3x, \quad f'(x) = 3\cos 3x, \quad f''(x) = -9\sin 3x,$$
$$f^{(3)}(x) = -27\cos 3x, \quad f^{(4)}(x) = 81\sin 3x, \quad f^{(5)}(x) = 243\cos 3x.$$

So $f(0) = 0$, $f'(0) = 3$, $f''(0) = 0$, $f^{(3)}(0) = -27$, $f^{(4)}(0) = 0$, $f^{(5)}(0) = 243$. Using the Maclaurin formula,

$$\sin 3x = 0 + 3x - 0 - \frac{27}{6}x^3 + 0 + \frac{243}{120}x^5 + \cdots, \quad \therefore \sin 3x \approx 3x - \frac{9}{2}x^3 + 0 + \frac{81}{40}x^5, \quad \text{valid for all } x.$$

4 c)

$$\begin{aligned} f(x) &= \cos\frac{x}{2}, \\ f'(x) &= -\frac{1}{2}\sin\frac{x}{2}, \\ f''(x) &= -\frac{1}{4}\cos\frac{x}{2}, \\ f^{(3)}(x) &= \frac{1}{8}\sin\frac{x}{2}, \\ f^{(4)}(x) &= \frac{1}{16}\cos\frac{x}{2}. \end{aligned}$$

So $f(0) = 1$, $f'(0) = 0$, $f''(0) = -\frac{1}{4}$, $f^{(3)}(0) = 0$, $f^{(4)}(0) = \frac{1}{16}$. Using Maclaurin,

$$\cos\frac{x}{2} = 1 + 0 + \frac{-1/4}{2}x^2 + 0 + \frac{1/16}{24}x^4 + \cdots, \quad \therefore \cos\frac{x}{2} \approx 1 - \frac{1}{8}x^2 + \frac{1}{384}x^4, \quad \text{valid for all } x.$$

4 d)

$$\begin{aligned} f(x) &= \ln(1+4x), \\ f'(x) &= 4(1+4x)^{-1}, \\ f''(x) &= -16(1+4x)^{-2}, \\ f^{(3)}(x) &= 128(1+4x)^{-3}. \end{aligned}$$

So $f(0) = 0$, $f'(0) = 4$, $f''(0) = -16$, $f^{(3)}(0) = 128$. Using the Maclaurin formula, $\ln(1+4x) = 0 + 4x - \frac{16}{2}x^2 + \frac{128}{6}x^3 + \cdots$.

$$\therefore \ln(1+4x) \approx 4x - 8x^2 + \frac{64}{3}x^3.$$

Validity: $-1 < 4x \leq 1 \Rightarrow -\frac{1}{4} < x \leq \frac{1}{4}$; the expansion is valid for $-\frac{1}{4} < x \leq \frac{1}{4}$.

4 e)

$$f(x) = \ln(x+2), \quad f'(x) = (x+2)^{-1}, \quad f''(x) = -(x+2)^{-2}.$$

So $f(0) = \ln 2$, $f'(0) = \frac{1}{2}$, $f''(0) = -\frac{1}{4}$. Using the Maclaurin formula,

$$\ln(x+2) = \ln 2 + \frac{1}{2}x + \frac{-1/4}{2!}x^2 + \cdots, \quad \therefore \ln(x+2) \approx \ln 2 + \frac{1}{2}x - \frac{1}{8}x^2.$$

Since $x + 2 = 2(1 + \frac{x}{2})$, validity is $-1 < \frac{x}{2} \leq 1 \Rightarrow -2 < x \leq 2$; the expansion is valid for $-2 < x \leq 2$.

4 f)

$$f(x) = (1+2x)^{1/4}, \quad f'(x) = \frac{1}{2}(1+2x)^{-3/4}, \quad f''(x) = -\frac{3}{4}(1+2x)^{-7/4}.$$

So $f(0) = 1$, $f'(0) = \frac{1}{2}$, $f''(0) = -\frac{3}{4}$. Using the Maclaurin formula,

$$(1+2x)^{1/4} = 1 + \frac{1}{2}x - \frac{3/4}{2!}x^2 + \cdots, \quad \therefore (1+2x)^{1/4} \approx 1 + \frac{1}{2}x - \frac{3}{8}x^2, \quad \text{valid for } |x| < \frac{1}{2}.$$

4 g)

$$f(x) = (1-2x)^{-3}, \quad f'(x) = 6(1-2x)^{-4}, \quad f''(x) = 48(1-2x)^{-5}.$$

So $f(0) = 1$, $f'(0) = 6$, $f''(0) = 48$. Using the Maclaurin formula,

$$(1-2x)^{-3} = 1 + 6x + \frac{48}{2!}x^2 + \cdots.$$

Validity: $|2x| < 1 \Rightarrow |x| < \frac{1}{2}$. Therefore $\dfrac{1}{(1-2x)^3} \approx 1 + 6x + 24x^2$, valid for $|x| < \frac{1}{2}$.

5 a)

$$\begin{aligned} f(x) &= \cosh x, \\ f'(x) &= \sinh x, \\ f''(x) &= \cosh x, \\ f^{(3)}(x) &= \sinh x, \\ f^{(4)}(x) &= \cosh x. \end{aligned}$$

So $f(0) = 1$, $f'(0) = 0$, $f''(0) = 1$, $f^{(3)}(0) = 0$, $f^{(4)}(0) = 1$. Using the Maclaurin formula,

$$\cosh x = 1 + 0 + \frac{1}{2}x^2 + 0 + \frac{1}{24}x^4 + \cdots, \quad \therefore \cosh x \approx 1 + \frac{1}{2}x^2 + \frac{1}{24}x^4.$$

5 b)

$$\begin{aligned} f(x) &= \sinh x, \\ f'(x) &= \cosh x, \\ f''(x) &= \sinh x, \\ f^{(3)}(x) &= \cosh x, \\ f^{(4)}(x) &= \sinh x, \\ f^{(5)}(x) &= \cosh x. \end{aligned}$$

So $f(0) = 0$, $f'(0) = 1$, $f''(0) = 0$, $f^{(3)}(0) = 1$, $f^{(4)}(0) = 0$, $f^{(5)}(0) = 1$. Using the Maclaurin formula,

$$\sinh x = 0 + x + 0 + \frac{1}{6}x^3 + 0 + \frac{1}{120}x^5 + \cdots, \quad \therefore \sinh x \approx x + \frac{1}{6}x^3 + \frac{1}{120}x^5.$$

6 a)

$$f(x) = (1+4x)^{1/2}, \quad f'(x) = 2(1+4x)^{-1/2}, \quad f''(x) = -4(1+4x)^{-3/2}.$$

6 b)

So $f(0) = 1$, $f'(0) = 2$, $f''(0) = -4$. Using the Maclaurin formula, $(1+4x)^{1/2} = 1 + 2x + \frac{-4}{2!}x^2 + \cdots$, therefore $\sqrt{1+4x} \approx 1 + 2x - 2x^2$.

6 c)

The expansion is valid for $|x| < \frac{1}{4}$.

6 d)

$1 + 4x = 6 \Rightarrow x = \frac{5}{4}$, which is not valid for this expansion. $\sqrt{6} = \sqrt{4}(1 + \frac{1}{2}) = 2\sqrt{1+\frac{1}{2}}$, and so use $x = \frac{1}{8}$.

$$\sqrt{1+\frac{1}{2}} \approx 1 + 2\left(\frac{1}{8}\right) - 2\left(\frac{1}{8}\right)^2 = \frac{39}{32}.$$

And so $\sqrt{6} = 2\sqrt{\frac{3}{2}} \approx 2 \times \frac{39}{32} = \frac{39}{16}$, as req.

7 a)

$$f(x) = \ln\left(\frac{x+2}{x+1}\right) = \ln(x+2) - \ln(x+1), \quad f(0) = \ln 2.$$

$$f'(x) = \frac{1}{x+2} - \frac{1}{x+1}, \quad f'(0) = -\frac{1}{2}.$$

$$f''(x) = -\frac{1}{(x+2)^2} + \frac{1}{(x+1)^2}, \quad f''(0) = \frac{3}{4}.$$

Using the Maclaurin formula,

$$\ln\left(\frac{x+2}{x+1}\right) = \ln 2 - \frac{1}{2}x + \frac{3/4}{2!}x^2 + \cdots, \quad \therefore \ln\left(\frac{x+2}{x+1}\right) \approx \ln 2 - \frac{1}{2}x + \frac{3}{8}x^2.$$

7 b)

Consider $\ln(x+2) = \ln(2(1+\frac{x}{2})) = \ln 2 + \ln(1+\frac{x}{2})$. Its validity is $-1 < \frac{x}{2} \leq 1 \Rightarrow -2 < x \leq 2$. The expansion for $\ln(x+1)$ is valid for $-1 < x \leq 1$. Therefore $-1 < x \leq 1$.

8 a)

$$\begin{aligned}
\sin\left(x - \frac{\pi}{4}\right) &= \sin x \cos\frac{\pi}{4} - \cos x \sin\frac{\pi}{4} \\
&= \frac{\sqrt{2}}{2}\left(x - \frac{x^3}{3!} + \cdots\right) - \frac{\sqrt{2}}{2}\left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} + \cdots\right) \\
&= \frac{\sqrt{2}}{2}\left(-1 + x + \frac{1}{2}x^2 - \frac{1}{6}x^3 - \frac{1}{24}x^4 + \cdots\right), \quad \text{as req.}
\end{aligned}$$

8 b)

$$
\begin{aligned}
f(x) &= \sin(x-\frac{\pi}{4}), \quad f(0) = -\frac{\sqrt{2}}{2};\\
f'(x) &= \cos(x-\frac{\pi}{4}), \quad f'(0) = \frac{\sqrt{2}}{2};\\
f''(x) &= -\sin(x-\frac{\pi}{4}), \quad f''(0) = \frac{\sqrt{2}}{2};\\
f^{(3)}(x) &= -\cos(x-\frac{\pi}{4}), \quad f^{(3)}(0) = -\frac{\sqrt{2}}{2};\\
f^{(4)}(x) &= \sin(x-\frac{\pi}{4}), \quad f^{(4)}(0) = -\frac{\sqrt{2}}{2}.
\end{aligned}
$$

Using the Maclaurin formula,

$$
\begin{aligned}
\sin(x-\frac{\pi}{4}) &= -\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}x + \frac{\sqrt{2}/2}{2}x^2 - \frac{\sqrt{2}/2}{6}x^3\\
&\quad - \frac{\sqrt{2}/2}{24}x^4 + \cdots\\
&= \frac{\sqrt{2}}{2}(-1 + x + \frac{1}{2}x^2 - \frac{1}{6}x^3 - \frac{1}{24}x^4 + \cdots), \quad \text{as req.}
\end{aligned}
$$

9 a)

$$
\begin{aligned}
\cos(x+\frac{3\pi}{4}) &= \cos x \cos\frac{3\pi}{4} - \sin x \sin\frac{3\pi}{4}\\
&= -\frac{\sqrt{2}}{2}(1 - \frac{x^2}{2!} + \frac{x^4}{4!} + \cdots) - \frac{\sqrt{2}}{2}(x - \frac{x^3}{3!} + \cdots)\\
&= \frac{\sqrt{2}}{2}(-1 + \frac{x^2}{2} - \frac{x^4}{24} + \cdots) - \frac{\sqrt{2}}{2}(x - \frac{x^3}{6} + \cdots)\\
&= \frac{\sqrt{2}}{2}(-1 - x + \frac{1}{2}x^2 + \frac{1}{6}x^3 - \frac{1}{24}x^4 + \cdots), \quad \text{as req.}
\end{aligned}
$$

9 b)

$$
\begin{aligned}
f(x) &= \cos(x+\frac{3\pi}{4}), \quad f(0) = -\frac{\sqrt{2}}{2};\\
f'(x) &= -\sin(x+\frac{3\pi}{4}), \quad f'(0) = -\frac{\sqrt{2}}{2};\\
f''(x) &= -\cos(x+\frac{3\pi}{4}), \quad f''(0) = \frac{\sqrt{2}}{2};\\
f^{(3)}(x) &= \sin(x+\frac{3\pi}{4}), \quad f^{(3)}(0) = \frac{\sqrt{2}}{2};\\
f^{(4)}(x) &= \cos(x+\frac{3\pi}{4}), \quad f^{(4)}(0) = -\frac{\sqrt{2}}{2}.
\end{aligned}
$$

Using the Maclaurin formula,

$$
\cos(x+\frac{3\pi}{4}) = -\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}x + \frac{\sqrt{2}/2}{2!}x^2
$$

$$+\frac{\sqrt{2}/2}{3!}x^3-\frac{\sqrt{2}/2}{4!}x^4+\cdots$$
$$=\frac{\sqrt{2}}{2}(-1-x+\frac{1}{2}x^2+\frac{1}{6}x^3-\frac{1}{24}x^4+\cdots),\quad \text{as req.}$$

10 a)

$$f(x)=\sin 2x,\ f'(x)=2\cos 2x,\ f''(x)=-4\sin 2x,\ f^{(3)}(x)=-8\cos 2x.$$

So $f(0)=0$, $f'(0)=2$, $f''(0)=0$, $f^{(3)}(0)=-8$. Using the Maclaurin formula, $\sin 2x=0+2x+0-\frac{8}{3!}x^3+\cdots$, so $\sin 2x\approx 2x-\frac{4}{3}x^3$.

10 b)

$$f(x)=\cos 2x,\ f'(x)=-2\sin 2x,\ f''(x)=-4\cos 2x,\ f^{(3)}(x)=8\sin 2x,\ f^{(4)}(x)=16\cos 2x.$$

So $f(0)=1$, $f'(0)=0$, $f''(0)=-4$, $f^{(3)}(0)=0$, $f^{(4)}(0)=16$. Using the Maclaurin formula, $\cos 2x=1+0-\frac{4}{2!}x^2+0+\frac{16}{4!}x^4+\cdots$, so $\cos 2x\approx 1-2x^2+\frac{2}{3}x^4$.

10 c)

$$\begin{aligned}\cos(2x+\frac{\pi}{6})&=\cos 2x\cos\frac{\pi}{6}-\sin 2x\sin\frac{\pi}{6}\\&\approx(1-2x^2+\frac{2}{3}x^4)\frac{\sqrt{3}}{2}-(2x-\frac{4}{3}x^3)\frac{1}{2}\\&=\frac{\sqrt{3}}{2}-x-\sqrt{3}x^2+\frac{2}{3}x^3+\frac{\sqrt{3}}{3}x^4,\quad\text{as req.}\end{aligned}$$

11 a)

$$\begin{aligned}f(x)&=\sin(\frac{\pi}{3}+x),\quad f(0)=\frac{\sqrt{3}}{2};\\f'(x)&=\cos(\frac{\pi}{3}+x),\quad f'(0)=\frac{1}{2};\\f''(x)&=-\sin(\frac{\pi}{3}+x),\quad f''(0)=-\frac{\sqrt{3}}{2}.\end{aligned}$$

Using the Maclaurin formula, $\sin(\frac{\pi}{3}+x)=\frac{\sqrt{3}}{2}+\frac{1}{2}x+\frac{-\sqrt{3}/2}{2!}x^2+\cdots\approx\frac{\sqrt{3}}{2}+\frac{1}{2}x-\frac{\sqrt{3}}{4}x^2$.

11 b)

$$\begin{aligned}\int_0^h x^2\sin(\frac{\pi}{3}+x)\,dx&\approx\int_0^h x^2(\frac{\sqrt{3}}{2}+\frac{1}{2}x-\frac{\sqrt{3}}{4}x^2)\,dx\\&=\int_0^h(\frac{\sqrt{3}}{2}x^2+\frac{1}{2}x^3-\frac{\sqrt{3}}{4}x^4)\,dx\\&=[\frac{\sqrt{3}}{6}x^3+\frac{1}{8}x^4-\frac{\sqrt{3}}{20}x^5]_0^h\end{aligned}$$

$$= (\frac{\sqrt{3}}{6}h^3 + \frac{1}{8}h^4 - \frac{\sqrt{3}}{20}h^5) - (0)$$
$$= \frac{\sqrt{3}h^3}{6} + \frac{h^4}{8} - \frac{\sqrt{3}h^5}{20}, \quad \text{as req.}$$

12)

$$f(x) = \cos 2x, \quad f'(x) = -2\sin 2x, \quad f''(x) = -4\cos 2x, \quad f^{(3)}(x) = 8\sin 2x,$$
$$f^{(4)}(x) = 16\cos 2x, \quad f^{(5)}(x) = -32\sin 2x, \quad f^{(6)}(x) = -64\cos 2x.$$

So $f(0) = 1$, $f'(0) = 0$, $f''(0) = -4$, $f^{(3)}(0) = 0$, $f^{(4)}(0) = 16$, $f^{(5)}(0) = 0$, $f^{(6)}(0) = -64$. Using

$$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f^{(3)}(0)}{3!}x^3 + \frac{f^{(4)}(0)}{4!}x^4 + \frac{f^{(5)}(0)}{5!}x^5 + \cdots + \frac{f^{(r)}(0)}{r!}x^r,$$
$$\cos 2x = 1 + 0 + \frac{-4}{2}x^2 + 0 + \frac{16}{24}x^4 + 0 + \frac{-64}{720}x^6 + \cdots \approx 1 - 2x^2 + \frac{2}{3}x^4 - \frac{4}{45}x^6.$$

Since $\cos(2A) = 2\cos^2 A - 1$, then $\cos^2 A = \frac{1}{2} + \frac{1}{2}\cos(2A)$, and so

$$\cos^2 x \approx \frac{1}{2} + \frac{1}{2}(1 - 2x^2 + \frac{2}{3}x^4 - \frac{4}{45}x^6) = 1 - x^2 + \frac{1}{3}x^4 - \frac{2}{45}x^6.$$

13 a)

Since $\cos(2A) = 1 - 2\sin^2 A$, then $\sin^2 A = \frac{1}{2} - \frac{1}{2}\cos(2A)$. So

$$\sin^2(\frac{x}{2}) = \frac{1}{2} - \frac{1}{2}\cos x \approx \frac{1}{2} - \frac{1}{2}(1 - \frac{1}{2}x^2 + \frac{1}{24}x^4) = \frac{1}{4}x^2 - \frac{1}{48}x^4.$$

13 b)

$$\int_{\pi/3}^{\pi/2} \frac{1}{x}\sin^2(\frac{x}{2})\,dx \approx \int_{\pi/3}^{\pi/2} \frac{1}{x}(\frac{1}{4}x^2 - \frac{1}{48}x^4)\,dx$$
$$= \int_{\pi/3}^{\pi/2} (\frac{1}{4}x - \frac{1}{48}x^3)\,dx$$
$$= [\frac{1}{8}x^2 - \frac{1}{192}x^4]_{\pi/3}^{\pi/2}$$
$$= (\frac{1}{8}(\frac{\pi}{2})^2 - \frac{1}{192}(\frac{\pi}{2})^4) - (\frac{1}{8}(\frac{\pi}{3})^2 - \frac{1}{192}(\frac{\pi}{3})^4) = 0.15 \text{ (2 DP)}.$$

14 a)

$$f(x) = \cos 2x, \quad f'(x) = -2\sin 2x, \quad f''(x) = -4\cos 2x, \quad f^{(3)}(x) = 8\sin 2x,$$
$$f^{(4)}(x) = 16\cos 2x, \quad f^{(5)}(x) = -32\sin 2x, \quad f^{(6)}(x) = -64\cos 2x.$$

So $f(0) = 1$, $f'(0) = 0$, $f''(0) = -4$, $f^{(3)}(0) = 0$, $f^{(4)}(0) = 16$, $f^{(5)}(0) = 0$, $f^{(6)}(0) = -64$. Using the Maclaurin formula,

$$\cos 2x = 1 + 0 + \frac{-4}{2}x^2 + 0 + \frac{16}{24}x^4 + 0 + \frac{-64}{720}x^6 + \cdots, \quad \text{giving } \cos 2x \approx 1 - 2x^2 + \frac{2}{3}x^4 - \frac{4}{45}x^6.$$

Since $\sin^2 x = \frac{1}{2} - \frac{1}{2}\cos 2x$, then

$$\sin^2 x \approx \frac{1}{2} - \frac{1}{2}(1 - 2x^2 + \frac{2}{3}x^4 - \frac{4}{45}x^6), \quad \therefore \sin^2 x \approx x^2 - \frac{1}{3}x^4 + \frac{2}{45}x^6.$$

14 b)

$$\begin{aligned}\int_0^{\pi/2} \sin^2 x\, dx &= \int_0^{\pi/2} (\frac{1}{2} - \frac{1}{2}\cos 2x)\, dx \\ &\approx \int_0^{\pi/2} \left(\frac{1}{2} - \frac{1}{2}(1 - 2x^2 + \frac{2}{3}x^4 - \frac{4}{45}x^6)\right) dx \\ &= \int_0^{\pi/2} (x^2 - \frac{1}{3}x^4 + \frac{2}{45}x^6) dx \\ &= [\frac{1}{3}x^3 - \frac{1}{15}x^5 + \frac{2}{315}x^7]_0^{\pi/2} \\ &= (\frac{1}{3}(\frac{\pi}{2})^3 - \frac{1}{15}(\frac{\pi}{2})^5 + \frac{2}{315}(\frac{\pi}{2})^7) - (0) \\ &= \frac{\pi^3}{24} - \frac{\pi^5}{480} + \frac{\pi^7}{20160} = 0.804 \text{ (3 SF)}.\end{aligned}$$

14 c)

Since $\cos(2A) = 1 - 2\sin^2 A$, then $\sin^2 A = \frac{1}{2} - \frac{1}{2}\cos(2A)$.

$$\int_0^{\pi/2} \sin^2 x\, dx = \int_0^{\pi/2} (\frac{1}{2} - \frac{1}{2}\cos 2x) dx = [\frac{1}{2}x - \frac{1}{4}\sin 2x]_0^{\pi/2} = (\frac{1}{4}\pi - 0) - (0 - 0) = \frac{\pi}{4}.$$

14 d)

$$\frac{\frac{\pi}{4} - (\frac{\pi^3}{24} - \frac{\pi^5}{480} + \frac{\pi^7}{20160})}{\frac{\pi}{4}} \times 100 = -2.4\% \text{ (2 SF)}.$$

15 a)

Since $1 - x + \sin x > 0$, then $\sin x > x - 1$. But $\sin x$ satisfies $-1 \leq \sin x \leq 1$, so $\sin x \leq 1 \Rightarrow 1 - x + \sin x \leq 2 - x$. If $x \geq 2$, then $2 - x \leq 0$, so $1 - x + \sin x \leq 0$ and $f(x)$ is not defined. So $x < 2$. Therefore k can never be greater than 2, as req.

15 b)

$$f(x) = \ln(1 - x + \sin x), \qquad f'(x) = \frac{-1 + \cos x}{1 - x + \sin x}.$$

15 c)

$\cos x \approx 1 - \frac{1}{2}x^2$ and $\sin x \approx x$. So

$$f'(x) = \frac{-1 + \cos x}{1 - x + \sin x} \approx \frac{-1 + 1 - \frac{1}{2}x^2}{1 - x + x} = -\frac{1}{2}x^2.$$

Therefore $f'(x) \approx -\frac{1}{2}x^2 \Rightarrow f(x) \approx -\frac{1}{6}x^3 + c$. But $f(0) = \ln(1 - 0 + \sin 0) = 0$, so $c = 0$. Hence $f(x) \approx -\frac{1}{6}x^3$.

16 a)

Since $\cosh 2A = 2\cosh^2 A - 1$, then $\cosh^2 A = \frac{1}{2} + \frac{1}{2}\cosh 2A$.

$$f(x) = \cosh^2 x = \frac{1}{2} + \frac{1}{2}\cosh 2x, \qquad f'(x) = \sinh 2x, \qquad f''(x) = 2\cosh 2x.$$

16 b)

Every odd power of the Maclaurin series includes $\sinh 0$, which is zero.

16 c)

$$f^{(n)}(0) = 2^{n-1}\cosh 0 = 2^{n-1}, \qquad \frac{f^{(n)}(0)}{n!} = \frac{2^{n-1}}{n!}.$$

17 a)

$$\begin{aligned} f(x) &= \sinh x, \\ f'(x) &= \cosh x, \\ f''(x) &= \sinh x, \\ f^{(3)}(x) &= \cosh x, \\ f^{(4)}(x) &= \sinh x, \\ f^{(5)}(x) &= \cosh x. \end{aligned}$$

So $f(0) = 0$, $f'(0) = 1$, $f''(0) = 0$, $f^{(3)}(0) = 1$, $f^{(4)}(0) = 0$, $f^{(5)}(0) = 1$. Using the Maclaurin formula,

$$\sinh x = 0 + x + 0 + \frac{1}{6}x^3 + 0 + \frac{1}{120}x^5 + \cdots, \quad \therefore \sinh x \approx x + \frac{1}{6}x^3 + \frac{1}{120}x^5.$$

17 b)

$$\begin{aligned} \sinh^n x = (\sinh x)^n &\approx (x + \frac{1}{6}x^3 + \frac{1}{120}x^5)^n \\ &= (x(1 + \frac{1}{6}x^2 + \frac{1}{120}x^4))^n = x^n(1 + \frac{1}{6}x^2 + \frac{1}{120}x^4)^n \\ &= x^n\left[1 + n(\frac{1}{6}x^2 + \frac{1}{120}x^4) + \frac{n(n-1)}{2}(\frac{1}{6}x^2 + \frac{1}{120}x^4)^2 + \cdots\right] \\ &= x^n\left[1 + n(\frac{1}{6}x^2 + \frac{1}{120}x^4) + \frac{n(n-1)}{2}(\frac{1}{36}x^4 + \cdots) + \cdots\right] \\ &= x^n + \frac{1}{6}nx^{n+2} + \left(\frac{n}{120} + \frac{n(n-1)}{72}\right)x^{n+4} + \cdots \\ &\approx x^n + \frac{1}{6}nx^{n+2} + \frac{1}{360}(n(5n-2))x^{n+4}. \end{aligned}$$

18 a)

$$f(x) = \ln(1+\cos x), \quad f(0) = \ln 2, \quad f'(x) = \frac{-\sin x}{1+\cos x}, \quad f'(0) = 0.$$

$$f''(x) = \frac{-\cos x(1+\cos x) - \sin^2 x}{(1+\cos x)^2}, \quad f''(0) = -\frac{1}{2}.$$

Using the Maclaurin formula,

$$\ln(1+\cos x) = \ln 2 + 0 + \frac{-1/2}{2!}x^2 + \cdots, \quad \therefore \ln(1+\cos x) \approx \ln 2 - \frac{1}{4}x^2.$$

18 b)

$$\ln(1+\cos x) = 0 \Rightarrow 1+\cos x = 1 \Rightarrow x = \frac{\pi}{2} \quad \text{(choosing the smallest possible value of } x\text{)}.$$

$$\ln 2 - \frac{1}{4}x^2 = 0 \Rightarrow x = 2\sqrt{\ln 2}.$$

So $\frac{\pi}{2} \approx 2\sqrt{\ln 2} \Rightarrow \pi \approx 4\sqrt{\ln 2}$. But since $\ln 2 = \ln(1+1) \approx 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots$, then using $\ln 2 \approx 1 - \frac{1}{2} = \frac{1}{2}$ gives

$$\pi \approx 4\sqrt{\frac{1}{2}} = \frac{4}{\sqrt{2}} = 2\sqrt{2}, \quad \text{as req.}$$

19)

$$\begin{aligned} f(x) &= \sec x, \\ f(0) &= 1, \\ f'(x) &= \sec x \tan x, \\ f'(0) &= 0, \\ f''(x) &= \sec^3 x + \sec x \tan^2 x, \\ f''(0) &= 1. \end{aligned}$$

Using the Maclaurin formula, $\sec x = 1 + 0 + \frac{1}{2}x^2 + \cdots$, hence $\sec x \approx 1 + \frac{1}{2}x^2$.

20 a)

$$\begin{aligned} f(x) &= \cos x, \\ f'(x) &= -\sin x, \\ f''(x) &= -\cos x, \\ f^{(3)}(x) &= \sin x, \\ f^{(4)}(x) &= \cos x. \end{aligned}$$

So $f(0) = 1$, $f'(0) = 0$, $f''(0) = -1$, $f^{(3)}(0) = 0$, $f^{(4)}(0) = 1$. Using the Maclaurin formula,

$$\cos x = 1 + 0 - \frac{1}{2}x^2 + 0 + \frac{1}{24}x^4 + \cdots, \quad \therefore \cos x \approx 1 - \frac{1}{2}x^2 + \frac{1}{24}x^4, \quad \text{as req.}$$

20 b)

Since $\cos x \sec x = 1$, then

$$(1 - \frac{1}{2}x^2 + \frac{1}{24}x^4 + \cdots)(1 + ax^2 + bx^4 + \cdots) = 1,$$
$$1 - \frac{1}{2}x^2 + \frac{1}{24}x^4 + ax^2 - \frac{1}{2}ax^4 + bx^4 + \cdots = 1.$$

So $-\frac{1}{2} + a = 0 \Rightarrow a = \frac{1}{2}$. So $\frac{1}{24} - \frac{1}{2} \times \frac{1}{2} + b = 0 \Rightarrow b = \frac{5}{24}$.

21 a)

$$\begin{aligned} f(x) &= \sin x, \\ f'(x) &= \cos x, \\ f''(x) &= -\sin x, \\ f^{(3)}(x) &= -\cos x, \\ f^{(4)}(x) &= \sin x, \\ f^{(5)}(x) &= \cos x. \end{aligned}$$

So $f(0) = 0$, $f'(0) = 1$, $f''(0) = 0$, $f^{(3)}(0) = -1$, $f^{(4)}(0) = 0$, $f^{(5)}(0) = 1$. Using the Maclaurin formula,

$$\sin x = 0 + x + 0 - \frac{1}{6}x^3 + 0 + \frac{1}{120}x^5 + \cdots, \quad \therefore \sin x \approx x - \frac{1}{6}x^3 + \frac{1}{120}x^5.$$

21 b)

Since $\sin x \operatorname{cosec} x = 1$, then

$$(\frac{1}{x} + ax + bx^3 + \cdots)(x - \frac{1}{6}x^3 + \frac{1}{120}x^5 + \cdots) = 1,$$
$$1 - \frac{1}{6}x^2 + \frac{1}{120}x^4 + ax^2 - \frac{1}{6}ax^4 + bx^4 + \cdots = 1.$$

So $-\frac{1}{6} + a = 0 \Rightarrow a = \frac{1}{6}$. So $\frac{1}{120} - \frac{1}{6} \times \frac{1}{6} + b = 0 \Rightarrow b = \frac{7}{360}$. Therefore $a = \frac{1}{6}$ and $b = \frac{7}{360}$.

22 a)

$$f'(x) = 3x^2 \cos(x^3).$$

22 b)

Since $\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} + \cdots$, then

$$\cos(x^3) = 1 - \frac{(x^3)^2}{2!} + \frac{(x^3)^4}{4!} - \frac{(x^3)^6}{6!} + \cdots.$$

$$3x^2 \cos(x^3) \approx 3x^2(1 - \frac{1}{2}x^6 + \frac{1}{24}x^{12} - \frac{1}{720}x^{18}) = 3x^2 - \frac{3}{2}x^8 + \frac{1}{8}x^{14} - \frac{1}{240}x^{20}.$$

22 c)

$$\int (3x^2 - \frac{3}{2}x^8 + \frac{1}{8}x^{14} - \frac{1}{240}x^{20})dx = x^3 - \frac{1}{6}x^9 + \frac{1}{120}x^{15} - \frac{1}{5040}x^{21} + c.$$

22 d)

Since $f(0) = \sin 0 = 0$, then $c = 0$. Therefore

$$\sin(x^3) \approx x^3 - \frac{1}{6}x^9 + \frac{1}{120}x^{15} - \frac{1}{5040}x^{21}.$$

23 a)

$$f(x) = (1-x)^2e^x, \quad f'(x) = (1-x)^2e^x - 2(1-x)e^x = (1 - 2x + x^2 - 2 + 2x)e^x = (x^2-1)e^x.$$
$$f''(x) = (x^2-1)e^x + 2xe^x = (x^2 + 2x - 1)e^x, \quad \text{as req.}$$

23 b)

So $f(0) = 1$, $f'(0) = -1$, $f''(0) = -1$. Using the Maclaurin formula, $f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f^{(3)}(0)}{3!}x^3 + \frac{f^{(4)}(0)}{4!}x^4 + \frac{f^{(5)}(0)}{5!}x^5 + \cdots + \frac{f^{(r)}(0)}{r!}x^r$, hence

$$(1-x)^2e^x \approx 1 - x - \frac{1}{2}x^2.$$

24 a)

$$f(x) = (1-x)\sin x = \sin x - x\sin x, \quad f'(x) = (1-x)\cos x - \sin x.$$
$$f''(x) = -(1-x)\sin x - \cos x - \cos x = (x-1)\sin x - 2\cos x, \quad \text{as req.}$$

24 b)

So $f(0) = 0$, $f'(0) = 1$, $f''(0) = -2$. Using the Maclaurin formula,

$$(1-x)\sin x \approx x - x^2.$$

25 a)

$$f(x) = e^{2x}\sin x, \quad f(0) = 0, \quad f'(x) = e^{2x}\cos x + 2e^{2x}\sin x, \quad f'(0) = 1.$$

Therefore $f(0) = 0$ and $f'(0) = 1$.

25 b)

$$\begin{aligned} f''(x) &= -e^{2x}\sin x + 2e^{2x}\cos x + 2e^{2x}\cos x + 4e^{2x}\sin x \\ &= 4e^{2x}\cos x + 3e^{2x}\sin x, \\ 4f'(x) - 5f(x) &= 4(e^{2x}\cos x + 2e^{2x}\sin x) - 5(e^{2x}\sin x) \\ &= 4e^{2x}\cos x + 8e^{2x}\sin x - 5e^{2x}\sin x \\ &= 4e^{2x}\cos x + 3e^{2x}\sin x. \end{aligned}$$

Therefore $f''(x) = 4f'(x) - 5f(x)$, as req.

25 c)

$$f''(0) = 4f'(0) - 5f(0) = 4 - 5 \times 0 = 4, \quad f^{(3)}(0) = 4f''(0) - 5f'(0) = 4 \times 4 - 5 \times 1 = 11.$$

Using the Maclaurin formula,

$$e^{2x} \sin x = 0 + x + \frac{4}{2!}x^2 + \frac{11}{3!}x^3 + \cdots, \quad \therefore e^{2x} \sin x \approx x + 2x^2 + \frac{11}{6}x^3.$$

26 a)

$$f(x) = e^{-x} \cos x, \quad f(0) = 1, \quad f'(x) = -e^{-x} \sin x - e^{-x} \cos x, \quad f'(0) = -1.$$

Therefore $f(0) = 1$ and $f'(0) = -1$.

26 b)

$$\begin{aligned} f''(x) &= -e^{-x} \cos x + e^{-x} \sin x + e^{-x} \sin x + e^{-x} \cos x = 2e^{-x} \sin x, \\ -2f'(x) - 2f(x) &= -2(-e^{-x} \sin x - e^{-x} \cos x) - 2(e^{-x} \cos x) \\ &= 2e^{-x} \sin x + 2e^{-x} \cos x - 2e^{-x} \cos x = 2e^{-x} \sin x. \end{aligned}$$

Therefore $f''(x) = -2f'(x) - 2f(x)$, as req.

26 c)

$$f''(0) = -2f'(0) - 2f(0) = 2 - 2 = 0, \quad f^{(3)}(0) = -2f''(0) - 2f'(0) = 0 + 2 = 2.$$

Using the Maclaurin formula, $e^{-x} \cos x = 1 - x + 0 + \frac{2}{3!}x^3 + \cdots$, hence $e^{-x} \cos x \approx 1 - x + \frac{1}{3}x^3$.

27 a)

$$\begin{aligned} f(x) &= \sin x \cosh x, \\ f'(x) &= \sin x \sinh x + \cos x \cosh x, \\ f''(x) &= \sin x \cosh x + \cos x \sinh x + \cos x \sinh x - \sin x \cosh x = 2 \cos x \sinh x, \\ f^{(3)}(x) &= 2 \cos x \cosh x - 2 \sin x \sinh x, \\ f^{(4)}(x) &= 2 \cos x \sinh x - 2 \sin x \cosh x - 2 \sin x \cosh x - 2 \cos x \sinh x \\ &= -4 \sin x \cosh x. \end{aligned}$$

Therefore $f^{(4)}(x) = -4f(x)$, so $k = -4$.

27 b)

$$f(0) = 0, \quad f'(0) = 1, \quad f''(0) = 0, \quad f^{(3)}(0) = 2, \quad f^{(4)}(0) = 0.$$

Since $f^{(5)}(x) = -4f'(x)$, then $f^{(5)}(0) = -4 \times 1 = -4$. Using the Maclaurin formula,

$$\sin x \cosh x = 0 + x + 0 + \frac{2}{3!}x^3 + 0 + \frac{-4}{5!}x^5 + \cdots, \quad \therefore \sin x \cosh x \approx x + \frac{1}{3}x^3 - \frac{1}{30}x^5.$$

28 a)

$$f(x) = e^{e^x+1}, \quad f(0) = e^2, \quad f'(x) = e^x \times e^{e^x+1} = e^{e^x+1+x}, \quad f'(0) = e^2.$$
$$f''(x) = (e^x+1)e^{e^x+1+x}, \quad f''(0) = 2e^2.$$
$$f^{(3)}(x) = (e^x+1)^2 e^{e^x+1+x} + e^{e^x+1+2x}, \quad f^{(3)}(0) = 5e^2.$$

Using the Maclaurin formula,

$$e^{e^x+1} = e^2 + e^2x + \frac{2e^2}{2!}x^2 + \frac{5e^2}{3!}x^3 + \cdots, \quad \therefore e^{e^x+1} \approx e^2 + e^2x + e^2x^2 + \frac{5}{6}e^2x^3.$$

28 b)

$$e^x \approx 1 + x + \frac{1}{2}x^2 + \frac{1}{6}x^3 + \frac{1}{24}x^4 + \cdots.$$

So $e^x + 1 \approx 2 + x + \frac{1}{2}x^2 + \frac{1}{6}x^3 + \frac{1}{24}x^4 + \cdots$, and

$$e^{e^x+1} \approx e^{2+x+\frac{1}{2}x^2+\frac{1}{6}x^3+\frac{1}{24}x^4+\cdots} \approx e^2 e^{x+\frac{1}{2}x^2+\frac{1}{6}x^3}.$$

Consider

$$\begin{aligned} e^{x+\frac{1}{2}x^2+\frac{1}{6}x^3} &= 1 + (x + \frac{1}{2}x^2 + \frac{1}{6}x^3) + \frac{1}{2}(x + \frac{1}{2}x^2 + \frac{1}{6}x^3)^2 + \frac{1}{6}(x + \frac{1}{2}x^2 + \frac{1}{6}x^3)^3 + \cdots \\ &= 1 + x + \frac{1}{2}x^2 + \frac{1}{6}x^3 + \frac{1}{2}x^2 + \frac{1}{2}x^3 + \frac{1}{6}x^3 + \cdots \\ &\approx 1 + x + x^2 + \frac{5}{6}x^3. \end{aligned}$$

So $e^{e^x+1} \approx e^2 + e^2x + e^2x^2 + \frac{5}{6}e^2x^3$.

29 a)

$$f(x) = \frac{1}{1+e^{-x}}, \quad f'(x) = \frac{e^{-x}}{(1+e^{-x})^2}.$$

$$\begin{aligned} f''(x) &= \frac{-2(1+e^{-x})e^{-x} + (1+e^{-x})^2 e^{-x}}{(1+e^{-x})^4} \\ &= \frac{-2e^{-x} + (1+e^{-x})e^{-x}}{(1+e^{-x})^3} = \frac{-e^{-x} + e^{-2x}}{(1+e^{-x})^3}. \end{aligned}$$

$$\frac{-e^{-x} + e^{-2x}}{(1+e^{-x})^3} \times \frac{e^{3x}}{e^{3x}} = \frac{-e^{2x} + e^x}{(e^x+1)^3} = \frac{e^x(1-e^x)}{(e^x+1)^3}, \quad \text{as req.}$$

29 b)

$$f(0) = \frac{1}{2}, \quad f'(0) = \frac{1}{4}, \quad f''(0) = 0.$$

$$\frac{d}{dx}(e^x(1-e^x)) = -e^{2x} + e^x(1-e^x) = e^x - 2e^{2x}.$$

$$f^{(3)}(x) = \frac{(e^x - 2e^{2x})(e^x+1)^3 - 3e^{2x}(1-e^x)(e^x+1)^2}{(e^x+1)^6},$$

$$f^{(3)}(0) = -\frac{1}{8}.$$

Using the Maclaurin formula,

$$\frac{1}{1+e^{-x}} = \frac{1}{2} + \frac{1}{4}x + 0 + \frac{-1/8}{3!}x^3 + \cdots, \quad \therefore \frac{1}{1+e^{-x}} \approx \frac{1}{2} + \frac{1}{4}x - \frac{1}{48}x^3.$$

30 a)

$$\begin{aligned} f(x) &= \tan x, \\ f'(x) &= \sec^2 x = \frac{1}{\cos^2 x}, \\ f''(x) &= \frac{2\sin x \cos x}{\cos^4 x} = 2\tan x \sec^2 x. \end{aligned}$$

$$\begin{aligned} f^{(3)}(x) &= 2\tan x(2\tan x \sec^2 x) + (2\sec^2 x)\sec^2 x \\ &= 4\tan^2 x \sec^2 x + 2\sec^4 x = 2\sec^2 x(\sec^2 x + 2\tan^2 x), \quad \text{as req.} \end{aligned}$$

30 b)

$$f(0) = 0, \quad f'(0) = 1, \quad f''(0) = 0, \quad f^{(3)}(0) = 2.$$

Using the Maclaurin formula, $\tan x = 0 + x + 0 + \frac{2}{3!}x^3 + \cdots$, so $\tan x \approx x + \frac{1}{3}x^3$.

30 c)

$$\begin{aligned} \int_h^{3h} \frac{\tan x}{2x}\,dx &\approx \int_h^{3h} \frac{1}{2x}(x + \frac{1}{3}x^3)dx = \int_h^{3h} (\frac{1}{2} + \frac{1}{6}x^2)dx \\ &= [\frac{1}{2}x + \frac{1}{18}x^3]_h^{3h} = (\frac{1}{2}(3h) + \frac{1}{18}(3h)^3) - (\frac{1}{2}h + \frac{1}{18}h^3) \\ &= \frac{3}{2}h + \frac{3}{2}h^3 - \frac{1}{2}h - \frac{1}{18}h^3 = h + \frac{13}{9}h^3, \quad \text{as req.} \end{aligned}$$

31 a)

$$f(x) = \ln(\cos x), \qquad f'(x) = -\frac{\sin x}{\cos x} = -\tan x, \quad \text{as req.}$$

31 b)

$$\begin{aligned} \frac{d}{dx}(\sec^2 x) &= \frac{d}{dx}(\frac{1}{\cos^2 x}) = \frac{0 + 2\cos x \sin x}{\cos^4 x} \\ &= 2 \times \frac{\sin x}{\cos x} \times \frac{1}{\cos^2 x} = 2\tan x \sec^2 x, \quad \text{as req.} \end{aligned}$$

31 c)

$$\begin{aligned} f(x) &= \ln(\cos x),\ f(0) = 0, \\ f'(x) &= -\tan x,\ f'(0) = 0, \\ f''(x) &= -\sec^2 x = -\frac{1}{\cos^2 x},\ f''(0) = -1. \\ f^{(3)}(x) &= -2\tan x \sec^2 x,\ f^{(3)}(0) = 0, \\ f^{(4)}(x) &= -4\tan^2 x \sec^2 x - 2\sec^4 x,\ f^{(4)}(0) = -2. \end{aligned}$$

Using the Maclaurin formula,

$$\ln(\cos x) = 0 + 0 - \frac{1}{2}x^2 + 0 - \frac{2}{24}x^4 + \cdots, \quad \therefore \ln(\cos x) \approx -\frac{1}{2}x^2 - \frac{1}{12}x^4.$$

31 d)

$\cos\frac{\pi}{6} = \frac{\sqrt{3}}{2}$ and so use $x = \frac{\pi}{6}$.

$$\ln(\cos\frac{\pi}{6}) \approx -\frac{1}{2}(\frac{\pi}{6})^2 - \frac{1}{12}(\frac{\pi}{6})^4, \quad \therefore \ln\frac{\sqrt{3}}{2} \approx -\frac{\pi^2}{72} - \frac{\pi^4}{15552}.$$

31 e)

$$\frac{(-\frac{\pi^2}{72} - \frac{\pi^4}{15552}) - \ln(\frac{\sqrt{3}}{2})}{\ln(\frac{\sqrt{3}}{2})} \times 100 = -0.347\%.$$

32 a)

$$\frac{d}{dx}(\ln(\sin x)) = \frac{1}{\sin x} \times \cos x = \cot x.$$

32 b)

$$\sin x \approx x - \frac{1}{6}x^3 + \frac{1}{120}x^6, \qquad \ln(1+x) \approx x - \frac{1}{2}x^2 + \frac{1}{3}x^3.$$

$$\begin{aligned} \ln(\sin x) &\approx \ln(x - \frac{1}{6}x^3 + \frac{1}{120}x^6) \\ &= \ln(x(1 - \frac{1}{6}x^2 + \frac{1}{120}x^5)) = \ln x + \ln(1 - \frac{1}{6}x^2 + \frac{1}{120}x^4) \\ &\approx \ln x + (-\frac{1}{6}x^2 + \frac{1}{120}x^4) - \frac{1}{2}(-\frac{1}{6}x^2 + \frac{1}{120}x^4)^2 + \frac{1}{3}(-\frac{1}{6}x^2 + \frac{1}{120}x^4)^3 \\ &= \ln x + (-\frac{1}{6}x^2 + \frac{1}{120}x^4) - \frac{1}{2}(\frac{1}{36}x^4 - \cdots) + O(x^6) \\ &= \ln x - \frac{1}{6}x^2 - \frac{1}{180}x^4 + O(x^6). \end{aligned}$$

So $\ln(\sin x) = \ln x - \frac{1}{6}x^2 - \frac{1}{180}x^4$, and differentiating gives

$$\cot x \approx \frac{1}{x} - \frac{1}{3}x - \frac{1}{45}x^3.$$

33 a)

$$\begin{aligned}\frac{d}{dx}\left(\ln(\tan(\frac{x}{2}))\right) &= \frac{1}{\tan(x/2)} \times \sec^2(\frac{x}{2}) \times \frac{1}{2} \\ &= \frac{\sec^2(x/2)}{2\tan(x/2)} = \frac{\cos(x/2)}{2\sin(x/2)\cos^2(x/2)} \\ &= \frac{1}{2\sin(x/2)\cos(x/2)}.\end{aligned}$$

Since $2\sin(\frac{x}{2})\cos(\frac{x}{2}) = \sin x$, then $\dfrac{d}{dx}(\ln(\tan(\frac{x}{2}))) = \dfrac{1}{\sin x} = \operatorname{cosec} x$, as req.

33 b)

$$\tan(\frac{x}{2}) = \frac{x}{2} + \frac{(x/2)^3}{3} + \frac{2(x/2)^5}{15} = \frac{x}{2} + \frac{x^3}{24} + \frac{x^5}{240}.$$

$$\begin{aligned}\ln(\tan(\frac{x}{2})) &\approx \ln(\frac{x}{2} + \frac{x^3}{24} + \frac{x^5}{240}) \\ &= \ln(\frac{x}{2}(1 + \frac{x^2}{12} + \frac{x^4}{120})) = \ln(\frac{x}{2}) + \ln(1 + \frac{x^2}{12} + \frac{x^4}{120}) \\ &= \ln(\frac{x}{2}) + (\frac{x^2}{12} + \frac{x^4}{120}) - \frac{1}{2}(\frac{x^2}{12} + \frac{x^4}{120})^2 + \cdots \\ &\approx \ln(\frac{x}{2}) + \frac{x^2}{12} + \frac{7x^4}{1440}.\end{aligned}$$

Since $\dfrac{d}{dx}(\ln(\tan(\frac{x}{2}))) = \operatorname{cosec} x$, then

$$\operatorname{cosec} x \approx \frac{d}{dx}(\ln(\frac{x}{2}) + \frac{x^2}{12} + \frac{7x^4}{1440}) = \frac{1}{x} + \frac{x}{6} + \frac{7x^3}{360}.$$

34 a)

$$y = \arcsin x \Rightarrow \sin y = x, \qquad \cos y \frac{dy}{dx} = 1 \Rightarrow \frac{dy}{dx} = \frac{1}{\cos y}.$$

But since $\sin y = \frac{x}{1}$, then $\cos y = \sqrt{1 - x^2}$, so $\dfrac{dy}{dx} = \dfrac{1}{\sqrt{1 - x^2}}$. Therefore $f'(x) = \dfrac{1}{\sqrt{1 - x^2}}$, as req.

34 b)

$$\begin{aligned}f(x) = \arcsin x,\ f(0) = 0, \quad f'(x) &= \frac{1}{\sqrt{1 - x^2}},\ f'(0) = 1, \\ f''(x) = x(1 - x^2)^{-3/2} &= \frac{x}{(1 - x^2)^{3/2}},\ f''(0) = 0, \\ f^{(3)}(x) &= \frac{(1 - x^2)^{3/2} + 3x^2(1 - x^2)^{1/2}}{(1 - x^2)^3},\ f^{(3)}(0) = 1.\end{aligned}$$

Using the Maclaurin formula, $\arcsin x = 0 + x + 0 + \frac{1}{6}x^3 + \cdots$, so $\arcsin x \approx x + \frac{1}{6}x^3$.

35 a)

$$y = \arccos x \Rightarrow \cos y = x, \qquad -\sin y \frac{dy}{dx} = 1 \Rightarrow \frac{dy}{dx} = -\frac{1}{\sin y}.$$

But since $\cos y = \frac{x}{1}$, then $\sin y = \sqrt{1-x^2}$, so $\frac{dy}{dx} = -\frac{1}{\sqrt{1-x^2}}$. Therefore $f'(x) = -\frac{1}{\sqrt{1-x^2}}$, as req.

35 b)

$$f(x) = \arccos x,\ f(0) = \frac{\pi}{2}, \quad f'(x) = -\frac{1}{\sqrt{1-x^2}},\ f'(0) = -1,$$
$$f''(x) = -x(1-x^2)^{-3/2} = -\frac{x}{(1-x^2)^{3/2}},\ f''(0) = 0,$$
$$f^{(3)}(x) = \frac{-(1-x^2)^{3/2} + 3x^2(1-x^2)^{1/2}}{(1-x^2)^3},\ f^{(3)}(0) = -1.$$

Using the Maclaurin formula, $\arccos x = \frac{\pi}{2} - x + 0 - \frac{1}{6}x^3 + \cdots$, hence $\arccos x \approx \frac{\pi}{2} - x - \frac{1}{6}x^3$.

36 a)

$$y = \arctan x \Rightarrow \tan y = x, \qquad \sec^2 y \frac{dy}{dx} = 1 \Rightarrow \frac{dy}{dx} = \frac{1}{\sec^2 y} = \cos^2 y.$$

But since $\tan y = \frac{x}{1}$, then $\cos y = \frac{1}{\sqrt{1+x^2}}$, so $\frac{dy}{dx} = (\frac{1}{\sqrt{1+x^2}})^2$. Therefore $f'(x) = \frac{1}{1+x^2}$, as req.

36 b)

$$f(x) = \arctan x,\ f(0) = 0,$$
$$f'(x) = \frac{1}{1+x^2},\ f'(0) = 1,$$
$$f''(x) = \frac{-2x}{(1+x^2)^2},\ f''(0) = 0.$$
$$f^{(3)}(x) = \frac{-2(1+x^2)^2 + 8x^2(1+x^2)}{(1+x^2)^4} = \frac{-2(1+x^2)^2 + 8x^2}{(1+x^2)^3} = \frac{-2+6x^2}{(1+x^2)^3},$$
$$f^{(3)}(0) = -2.$$

Using the Maclaurin formula, $\arctan x = 0 + x + 0 - \frac{2}{6}x^3 + \cdots$, hence $\arctan x \approx x - \frac{1}{3}x^3$.

36 c)

Since $\arctan 1 = \frac{\pi}{4}$, then $\arctan 1 = 1 - \frac{1}{3} + \cdots$, so $\frac{\pi}{4} = 1 - \frac{1}{3} + \cdots$.

37 a)

$$\frac{1}{\sqrt{1+x^2}} = (1+x^2)^{-1/2} = 1 + (-\frac{1}{2})(x^2) + \frac{(-\frac{1}{2})(-\frac{3}{2})}{2!}(x^2)^2$$
$$+ \frac{(-\frac{1}{2})(-\frac{3}{2})(-\frac{5}{2})}{3!}(x^2)^3 + \cdots$$
$$\approx 1 - \frac{1}{2}x^2 + \frac{3}{8}x^4 - \frac{5}{16}x^6.$$

37 b)

Since $\dfrac{d}{dx}(\operatorname{arsinh} x) = \dfrac{1}{\sqrt{1+x^2}}$, then

$$\operatorname{arsinh} x = \int \frac{1}{\sqrt{1+x^2}}dx \approx \int (1 - \frac{1}{2}x^2 + \frac{3}{8}x^4 - \frac{5}{16}x^6)dx$$
$$\therefore \operatorname{arsinh} x \approx x - \frac{1}{6}x^3 + \frac{3}{40}x^5 - \frac{5}{112}x^7.$$

38)

Since $\dfrac{d}{dx}(\operatorname{artanh} x) = \dfrac{1}{1-x^2}$,

$$\begin{aligned}\frac{1}{1-x^2} &= (1+x^2)^{-1} = 1 + (-1)(-x^2) + \frac{(-1)(-2)}{2!}(-x^2)^2 + \frac{(-1)(-2)(-3)}{3!}(-x^2)^3 + \cdots \\ &= 1 + x^2 + x^4 + x^6 + \cdots .\end{aligned}$$

So

$$\operatorname{artanh} x = \int \frac{1}{1+x^2}dx \approx \int (1 + x^2 + x^4 + x^6)dx \Rightarrow \operatorname{artanh} x \approx x + \frac{1}{3}x^3 + \frac{1}{5}x^5 + \frac{1}{7}x^7, \quad \text{as req.}$$

39 a)

$$f(x) = \arccos x, \quad f(0) = \frac{\pi}{2}, \qquad f'(x) = -\frac{1}{\sqrt{1-x^2}}, \quad f'(0) = -1.$$
$$f''(x) = -x(1-x^2)^{-3/2} = -\frac{x}{(1-x^2)^{3/2}}, \quad f''(0) = 0.$$
$$f^{(3)}(x) = \frac{-(1-x^2)^{3/2} + 3x^2(1-x^2)^{1/2}}{(1-x^2)^3}, \quad f^{(3)}(0) = -1.$$

Using the Maclaurin formula, $\arccos x = \frac{\pi}{2} - x + 0 - \frac{1}{6}x^3 + \cdots$, hence $\arccos x \approx \frac{\pi}{2} - x - \frac{1}{6}x^3$.

39 b)

Since $\ln(1+x) \approx x - \frac{1}{2}x^2 + \frac{1}{3}x^3 + \cdots$,

$$\begin{aligned}(\arccos x)\ln(1+x) &\approx \left(\frac{\pi}{2} - x - \frac{1}{6}x^3\right)\left(x - \frac{1}{2}x^2 + \frac{1}{3}x^3\right) \\ &= \frac{\pi}{2}x - \frac{\pi}{4}x^2 + \frac{\pi}{6}x^3 - x^2 + \frac{1}{2}x^3 + \cdots \\ &= \frac{\pi}{2}x + \left(-1 - \frac{\pi}{4}\right)x^2 + \left(\frac{1}{2} + \frac{\pi}{6}\right)x^3 + \cdots .\end{aligned}$$

So the expansion up to and including the x^3 term is $\frac{\pi}{2}x + \left(-1 - \frac{\pi}{4}\right)x^2 + \frac{1}{6}(3+\pi)x^3$, as required.

40 a)

$$f(x) = \arctan(1+x), \quad f(0) = \frac{\pi}{4}, \qquad f'(x) = \frac{1}{1+(1+x)^2}, \quad f'(0) = \frac{1}{2}.$$
$$f''(x) = \frac{-2(1+x)}{(1+(1+x)^2)^2}, \quad f''(0) = -\frac{1}{2}.$$

Using the Maclaurin formula, $\arctan(1+x) = \frac{\pi}{4} + \frac{1}{2}x - \frac{1/2}{2!}x^2 + \cdots$, hence $\arctan(1+x) \approx \frac{\pi}{4} + \frac{1}{2}x - \frac{1}{4}x^2$.

40 b)

$$\begin{aligned}\int_{-h}^{h} x\arctan(1+x)\,dx &\approx \int_{-h}^{h} x\left(\frac{\pi}{4}+\frac{1}{2}x-\frac{1}{4}x^2\right)dx = \int_{-h}^{h}\left(\frac{\pi}{4}x+\frac{1}{2}x^2-\frac{1}{4}x^3\right)dx\\ &= \left[\frac{\pi}{8}x^2+\frac{1}{6}x^3-\frac{1}{16}x^4\right]_{-h}^{h}\\ &= \left(\frac{\pi}{8}h^2+\frac{1}{6}h^3-\frac{1}{16}h^4\right)-\left(\frac{\pi}{8}(-h)^2+\frac{1}{6}(-h)^3-\frac{1}{16}(-h)^4\right)\\ &= \frac{1}{6}h^3+\frac{1}{6}h^3=\frac{1}{3}h^3, \quad \text{as required.}\end{aligned}$$

41 a)

$$f(x)=\arcsin x, \quad f(0)=0, \qquad f'(x)=\frac{1}{\sqrt{1-x^2}}, \quad f'(0)=1.$$

$$f''(x)=x(1-x^2)^{-3/2}=\frac{x}{(1-x^2)^{3/2}}, \quad f''(0)=0.$$

$$f^{(3)}(x)=\frac{(1-x^2)^{3/2}+3x^2(1-x^2)^{1/2}}{(1-x^2)^3}, \quad f^{(3)}(0)=1.$$

Thus $\arcsin x = 0+x+0+\frac{1}{6}x^3+\cdots$, so $\arcsin x \approx x+\frac{1}{6}x^3$.

41 b)

$$\begin{aligned}\int_0^{1/2}\arcsin x\,dx &\approx \int_0^{1/2}\left(x+\frac{1}{6}x^3\right)dx\\ &\approx \left[\frac{1}{2}x^2+\frac{1}{24}x^4\right]_0^{1/2}=\left(\frac{1}{8}+\frac{1}{384}\right)-(0+0)\\ &= \frac{49}{384}=0.128 \text{ (3 SF)}.\end{aligned}$$

41 c)

$$\int \arcsin x\,dx = \int 1\times\arcsin x\,dx.$$

Use integration by parts: $u=\arcsin x$, $\frac{dv}{dx}=1$, $\frac{du}{dx}=\frac{1}{\sqrt{1-x^2}}$, $v=x$.

$$\therefore \int \arcsin x\,dx = x\arcsin x-\int\frac{x}{\sqrt{1-x^2}}\,dx.$$

For $\int\frac{x}{\sqrt{1-x^2}}\,dx$, use integration by substitution. Let $u=1-x^2$, so $\frac{du}{dx}=-2x$ and $dx=-\frac{1}{2x}du$.

$$\therefore \int\frac{x}{\sqrt{1-x^2}}\,dx=\int\frac{x}{\sqrt{u}}\left(-\frac{1}{2x}\right)du=-\frac{1}{2}\int u^{-1/2}du=-u^{1/2}+c=-\sqrt{1-x^2}+c.$$

So $\int\arcsin x\,dx = x\arcsin x+\sqrt{1-x^2}+c$, and

$$\begin{aligned}\int_0^{1/2}\arcsin x\,dx &\approx \int_0^{1/2}\left(x+\frac{1}{6}x^3\right)dx\\ &\approx \left[\frac{1}{2}x^2+\frac{1}{24}x^4\right]_0^{1/2}=\left(\frac{1}{8}+\frac{1}{384}\right)-(0+0)\\ &= \frac{49}{384}=0.128 \text{ (3 SF)}.\end{aligned}$$

41 d)

$$\frac{\frac{49}{384} - \left(\frac{\pi}{12} - 1 + \frac{\sqrt{3}}{2}\right)}{\frac{\pi}{12} - 1 + \frac{\sqrt{3}}{2}} \times 100 = -0.2\% \text{ (1 DP)}.$$

42 a)

$$f(x) = \arcsin\left(x + \frac{1}{3}\right), \quad f(0) = \arcsin\frac{1}{3}.$$
$$f'(x) = \frac{1}{\sqrt{1-(x+1/3)^2}}, \quad f'(0) = \frac{3\sqrt{2}}{4}.$$
$$f''(x) = \left(x + \frac{1}{3}\right)\left(1 - \left(x + \frac{1}{3}\right)^2\right)^{-3/2},$$
$$f''(0) = \frac{9\sqrt{2}}{32}.$$

Hence $\arcsin(x + \frac{1}{3}) = \arcsin\frac{1}{3} + \frac{3\sqrt{2}}{4}x + \frac{(9\sqrt{2}/32)}{2!}x^2 + \cdots$, and

$$\arcsin\left(x + \frac{1}{3}\right) \approx \arcsin\frac{1}{3} + \frac{3\sqrt{2}}{4}x + \frac{9\sqrt{2}}{64}x^2.$$

42 b)

$$\begin{aligned}\int_0^{0.05} \arcsin\left(x + \frac{1}{3}\right) dx &\approx \int_0^{0.05} \left(\arcsin\frac{1}{3} + \frac{3\sqrt{2}}{4}x + \frac{9\sqrt{2}}{64}x^2\right) dx \\ &= \left[\arcsin\frac{1}{3}x + \frac{3\sqrt{2}}{8}x^2 + \frac{3\sqrt{2}}{64}x^3\right]_0^{0.05} \\ &= \left(\arcsin\frac{1}{3}(0.05) + \frac{3\sqrt{2}}{8}(0.05)^2 + \frac{3\sqrt{2}}{64}(0.05)^3\right) - (0) = 0.0183 \text{ (3 SF)}.\end{aligned}$$

43 a)

$$f(x) = \arccos(2x), \quad f'(x) = -\frac{2}{\sqrt{1-4x^2}}.$$

43 b)

Use the binomial series:

$$\begin{aligned}-2(1-4x^2)^{-1/2} &= -2\left(1 + \left(-\frac{1}{2}\right)(-4x^2) + \frac{(-1/2)(-3/2)}{2!}(-4x^2)^2 + \cdots\right) \\ &= -2 - 4x^2 - 12x^4 + \cdots.\end{aligned}$$

$$\int -\frac{2}{\sqrt{1-4x^2}}dx = \int \arccos(2x)\,dx.$$

$$\therefore \int -\frac{2}{\sqrt{1-4x^2}}dx \approx \int(-2 - 4x^2 - 12x^4)dx = -2x - \frac{4}{3}x^3 - \frac{12}{5}x^5 + c.$$

Since $\arccos(0) = \frac{\pi}{2}$, $\frac{\pi}{2} = 0 + c$, so $c = \frac{\pi}{2}$. Thus $\arccos(2x) \approx \frac{\pi}{2} - 2x - \frac{4}{3}x^3 - \frac{12}{5}x^5$.

44 a)

$$f(x) = \arcsin(2x), \quad f(0) = 0, \qquad f'(x) = \frac{2}{\sqrt{1-4x^2}} = 2(1-4x^2)^{-1/2}, \quad f'(0) = 2.$$
$$f''(x) = 8x(1-4x^2)^{-3/2}, \quad f''(0) = 0.$$
$$f^{(3)}(x) = 64x^2(1-4x^2)^{-5/2} + 8(1-4x^2)^{-3/2}, \quad f^{(3)}(0) = 8.$$

So $\arcsin(2x) = 0 + 2x + 0 + \frac{8}{3!}x^3 + \cdots$, hence $\arcsin(2x) \approx 2x + \frac{4}{3}x^3$.

44 b)

When $x = \frac{1}{4}$, $\arcsin(2 \times \frac{1}{4}) = \frac{\pi}{6}$. So

$$\frac{\pi}{6} \approx 2 \times \frac{1}{4} + \frac{4}{3}\left(\frac{1}{4}\right)^3 = \frac{1}{2} + \frac{1}{48}, \qquad \therefore \pi \approx 6\left(\frac{1}{2} + \frac{1}{48}\right), \quad \text{as required.}$$

44 c)

$$\frac{6(\frac{1}{2} + \frac{1}{48}) - \pi}{\pi} \times 100 = -0.528\%.$$

45 a)

$$\frac{d}{dx}(\operatorname{artanh} x) = \frac{1}{1-x^2}, \quad \therefore \frac{d}{dx}(\operatorname{artanh} u) = \frac{du}{dx}\frac{1}{1-u^2}.$$
$$u = \frac{1-x}{3+x}, \quad \frac{du}{dx} = \frac{-(3+x)-(1-x)}{(3+x)^2} = -\frac{4}{(3+x)^2}.$$

$$1 - u^2 = 1 - \left(\frac{1-x}{3+x}\right)^2 = \frac{(3+x)^2}{(3+x)^2} - \frac{(1-x)^2}{(3+x)^2}$$
$$= \frac{(3+x)^2 - (1-x)^2}{(3+x)^2} = \frac{9+6x+x^2-1+2x-x^2}{(3+x)^2} = \frac{8+8x}{(3+x)^2}.$$
$$\therefore f'(x) = \frac{-4/(3+x)^2}{(8+8x)/(3+x)^2} = -\frac{4}{8+8x} = -\frac{1}{2x+2}, \quad \text{as required.}$$

45 b)

$$f''(x) = \frac{2}{(2x+2)^2} = \frac{2}{4(x+1)^2} = \frac{1}{2(x+1)^2}.$$

45 c)

$$f(x) = \operatorname{artanh}\left(\frac{1-x}{3+x}\right),$$
$$f(0) = \operatorname{artanh}\frac{1}{3} = \frac{1}{2}\ln\left(\frac{1+1/3}{1-1/3}\right) = \frac{1}{2}\ln 2.$$
$$f'(x) = -\frac{1}{2x+2}, \quad f'(0) = -\frac{1}{2}, \qquad f''(x) = \frac{1}{2(x+1)^2}, \quad f''(0) = \frac{1}{2}.$$

So $\operatorname{artanh}\left(\frac{1-x}{3+x}\right) = \frac{1}{2}\ln 2 - \frac{1}{2}x + \frac{1/2}{2!}x^2 + \cdots$, hence $\operatorname{artanh}\left(\frac{1-x}{3+x}\right) \approx \frac{1}{2}\ln 2 - \frac{1}{2}x + \frac{1}{4}x^2$, as required.

46 a)

$$f(x) = \arctan(\cosh x), \quad \frac{d}{dx}(\arctan x) = \frac{1}{1+x^2}, \quad \therefore \frac{d}{dx}(\arctan u) = \frac{du}{dx}\frac{1}{1+u^2}.$$

$$u = \cosh x, \quad \frac{du}{dx} = \sinh x, \quad f'(x) = \frac{\sinh x}{1+\cosh^2 x}.$$

$$f''(x) = \frac{\cosh x(1+\cosh^2 x) - 2\sinh^2 x \cosh x}{(1+\cosh^2 x)^2} = \frac{\cosh x(1+\cosh^2 x - 2\sinh^2 x)}{(1+\cosh^2 x)^2}.$$

46 b)

$$f(x) = \arctan(\cosh x),$$

$$f(0) = \frac{\pi}{4}, \qquad f'(x) = \frac{\sinh x}{1+\cosh^2 x},$$

$$f'(0) = 0.$$

$$f''(x) = \frac{\cosh x(1+\cosh^2 x - 2\sinh^2 x)}{(1+\cosh^2 x)^2}, \quad f''(0) = \frac{1}{2}.$$

So $\arctan(\cosh x) = \frac{\pi}{4} + 0 + \frac{1/2}{2!}x^2 + \cdots$, hence $\arctan(\cosh x) \approx \frac{\pi}{4} + \frac{1}{4}x^2$.